

OBRESTAD, Norway

WMO: 014120

Lat: 58.6585N

Lon: 5.5546E

Elev: 26

StdP: 101.01

Time Zone: 1.00 (EUC)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-7.5	-5.5	-14.0	1.1	-5.3	-11.4	1.4	-3.8	20.3	3.7	18.4	4.4	3.7	70	0.832

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.1	21.4	16.5	19.8	15.7	18.6	15.4	17.6	20.2	16.7	19.0	15.9	18.0	5.7	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.4	11.7	18.9	15.6	11.1	18.1	14.9	10.6	17.2	49.5	20.3	47.0	19.0	44.8	18.1	21.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.3	16.0	14.3	-8.2	25.9	2.6	1.7	-10.1	27.2	-11.6	28.1	-13.1	29.1	-15.0	30.3		
(5)			-8.9	19.4	2.8	1.2	-10.9	20.3	-12.5	20.9	-14.1	21.6	-16.1	22.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.1	2.3	1.8	3.0	6.2	9.1	11.7	14.7	15.1	12.8	9.4	6.0	4.1
	DBStd	5.43	3.62	3.20	2.88	2.49	2.91	2.36	2.53	2.07	2.22	2.64	3.50	3.71
	HDD10.0	1213	237	229	218	117	51	7	0	0	4	43	122	184
	HDD18.3	3761	496	463	477	363	287	198	118	102	167	278	369	442
	CDD10.0	505	0	0	0	4	23	59	146	159	87	23	3	0
	CDD18.3	10	0	0	0	0	1	1	5	3	1	0	0	0
	CDH23.3	17	0	0	0	0	2	1	11	4	0	0	0	0
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0
Wind	WSAvg	7.0	8.2	7.2	6.5	6.2	6.5	6.2	6.0	6.4	7.1	7.5	7.6	8.0
Precipitation	PrecAvg	1536	158	122	103	85	72	79	103	148	157	177	171	161
	PrecMax	1921	285	263	242	153	142	155	216	287	261	332	311	303
	PrecMin	1039	4	30	14	35	11	20	32	20	41	59	47	4
	PrecStd	219	67	64	52	29	34	37	46	54	56	75	76	77
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.2	8.0	9.9	15.9	20.6	21.2	24.6	23.6	20.2	15.7	12.7	10.5
		MCWB	7.8	7.0	6.1	10.3	14.0	15.4	18.0	15.4	15.1	13.3	11.5	9.3
	2%	DB	8.2	7.0	7.6	13.3	18.0	18.8	21.9	21.0	18.1	14.7	11.9	9.7
		MCWB	7.0	6.0	5.5	8.5	13.4	14.8	17.5	16.1	14.9	12.8	10.8	8.5
	5%	DB	7.5	6.4	6.7	11.1	15.5	16.8	20.0	19.4	16.9	13.7	11.2	9.1
		MCWB	6.4	5.6	5.3	7.9	11.5	13.9	16.7	15.5	14.4	12.1	9.9	8.0
	10%	DB	7.0	5.9	6.1	9.6	13.2	15.2	18.4	18.4	15.9	12.9	10.4	8.4
		MCWB	5.9	5.1	5.0	6.9	10.4	13.0	16.0	15.3	13.9	11.4	9.1	7.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.2	7.2	7.1	10.9	16.3	16.6	19.2	18.3	16.2	14.1	11.9	9.7
		MCDB	8.8	7.7	8.7	14.8	20.2	19.3	23.2	20.8	18.4	15.2	12.5	10.1
	2%	WB	7.4	6.4	6.2	9.2	13.6	15.4	18.1	17.2	15.5	13.3	11.1	9.0
		MCDB	8.1	7.0	7.0	11.7	16.9	18.4	21.0	19.4	17.4	14.3	11.9	9.6
	5%	WB	6.8	6.0	5.7	8.4	11.9	14.1	17.2	16.5	14.9	12.6	10.2	8.4
		MCDB	7.6	6.4	6.4	10.4	14.4	16.5	19.7	18.5	16.5	13.5	11.0	9.1
	10%	WB	6.2	5.4	5.3	7.5	10.6	13.1	16.2	15.8	14.2	11.8	9.5	7.6
		MCDB	7.0	6.0	6.0	9.0	12.6	14.8	18.3	17.7	15.9	12.9	10.4	8.4
Mean Daily Temperature Range	5% DB	MDBR	3.6	3.5	3.8	4.4	4.2	4.0	3.9	4.1	4.0	4.1	3.8	3.7
		MCDBR	3.7	3.0	3.8	6.8	8.1	6.8	6.8	5.7	4.9	3.7	3.1	3.2
		MCWBR	3.9	2.9	3.2	4.2	4.8	4.4	4.2	3.8	3.4	3.3	3.3	3.6
	5% WB	MCDBR	3.6	2.8	3.6	5.6	7.4	5.9	6.3	4.7	4.3	3.4	3.3	3.1
		MCWBR	3.9	2.9	3.0	4.0	4.7	4.3	4.1	3.6	3.6	3.3	3.7	3.5
Clear-Sky Solar Irradiance	taub	0.314	0.336	0.354	0.385	0.372	0.366	0.379	0.381	0.363	0.343	0.317	0.292	
	taud	2.173	2.272	2.319	2.290	2.403	2.451	2.432	2.424	2.458	2.395	2.246	2.129	
	Ebn at Noon	484	677	783	818	860	871	848	817	773	669	484	377	
	Edh at Noon	50	72	91	111	106	102	102	96	80	64	46	37	
All-Sky Solar Radiation	RadAvg	0.30	0.91	2.14	3.85	5.23	5.62	5.02	3.88	2.50	1.19	0.42	0.19	
	RadStd	0.04	0.14	0.21	0.35	0.54	0.61	0.48	0.32	0.21	0.13	0.06	0.02	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (28 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	-116	N/A	N/A

Nomenclature: See separate page